JUNK MILL

CONVENTIONAL JUNK MILL

MILLING

GENERAL DESCRIPTION

THE WORKHORSE OF DOWNHOLE MILLING, USED TO MILL ALMOST ANYTHING IN THE HOLE INCLUDING CEMENT AND RUBBER PRODUCTS. IT IS THE ONLY TOOL ABLE TO MILL DRILL PIPE CEMENTED INSIDE AND OUT. IT IS ALSO USED TO MILL CAST-IRON BRIDGE PLUGS, AND TO MILL CASING THAT HAS BEEN MILLED TO THE POINT OF ROTATING, RUN ON A JOINT OF SLIGHTLY BENT DRILL PIPE. THREE DRESSING OPTIONS ARE LISTED: CONVENTIONAL, CONEBUSTER AND SUPER.

TECHNICAL CHARACTERISTICS

Cutting diameter (in.)	Top pin conn. (API Reg.)	Overall length: conventional and Conebuster (in.)	Overall length: Super (in.)	Fishing neck length (in.)	Fishing neck dia. (in.)	Wt (lb)
3-1/2 - 4-1/2	2-3/8	20	20	12	3	45
4-1/2 - 5-1/2	2-7/8	21	21	12	3-3/4	62
5-1/2 - 5-5/8	3-1/2	23	21	12	4-1/4	95
5-3/4 - 7-1/2	3-1/2	23	21	12	4-3/4	105
7-1/2 - 9	4-1/2	27	27	12	5-3/4	180
9-1/2 - 12-1/4	6-5/8	29	29	12	7-3/4	350
13 - 15	6-5/8 or 7-5/8	30	30	12	7-3/4 or 9-1/2	500
17 - 17-1/2	6-5/8 or 7-5/8	33	33	12	7-3/4 or 9-1/2	625
18-1/2 - 26	6-5/8 or 7-5/8	37	37	12	7-3/4 or 9-1/2	1,200

Dimensions in inches, weights in lb (approximate).

DESIGN FEATURES

- **Fishing neck:** allows retrieval of the mill if it is left in the hole.
- **Dressing options:** conventional, Conebuster or Super.
- **Inside casing:** mill head O.D. equal to the casing drift; wear pads of the same O.D. protect the casing; stabilizer of the same O.D. run directly above.
- **Bridge plugs:** mill O.D. about 1/8 in. under the plug size, to avoid skinning the casing.

OPERATING NOTES

- **Open hole:** mill O.D. about 1/8 in. under hole diameter; at least 10,000 lb of drill collars; junk sub directly above the mill.
- **Spudding:** frequent spudding pounds loose junk onto bottom for effective milling; a rise in torque shows junk alongside the mill.
- **Starting:** set down on the junk while turning, then bring weight up; never apply weight before rotating.

PERFORMANCE PARAMETERS

Weight on mill	4,000 - 10,000 lb
Start-up rotary speed	60 - 80 rpm
Rotary speed once milling	80 - 100 rpm, normal pump
Weight, second pass	4,000 - 6,000 lb
Surface speed	250 - 340 ft/min
Rotary speed	Surf. speed / (O.D. x 3.14)
Loose junk cycle	Mill 1-2 ft, lift 15-20 ft, settle
Stationary junk cycle	Each 3-5 ft, lift 10-15 ft, ream
Rough running	Cut speed/weight, raise slowly
Stabilizer spacing	60 - 90 ft above mill (vertical)

HOW TO ORDER

Please specify: 1. mill diameter 2. size and weight of the casing to run through, if available 3. top connection 4. dressing (conventional, Conebuster or Super).